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✓ Task

Perform a bootstrap analysis on the 'Emissions_Canada (1).csv' dataset to determine the 90% and 95% confidence intervals for the linear regression coefficient (β_1) of 'Engine Size' predicting 'CO2 Emissions', visualize the distribution of bootstrapped coefficients with the confidence intervals, and conclude on the significance of 'Engine Size' on 'CO2 Emissions'.

✓ Load and Prepare Data

Subtask:

Load the 'Emissions_Canada (1).csv' dataset into a pandas DataFrame. Select 'Engine Size' as the feature (X) and 'CO2 Emissions' as the target (y).

Reasoning: To address the subtask requirements, I'll first import the pandas library, then load the specified CSV file into a DataFrame. Following that, I'll extract the 'Engine Size' column into variable X and the 'CO2 Emissions' column into variable y.

Reasoning: The previous code failed because the column names 'Engine Size' and 'CO2 Emissions' were incorrect. I will correct them to 'Engine Size(L)' and 'CO2 Emissions(g/km)' based on the `df.head()` output.

```
import pandas as pd

df = pd.read_csv('/content/Emissions_Canada (1).csv')
print("DataFrame loaded successfully. Displaying the first 5 rows:")
print(df.head())

X = df[['Engine Size(L)']]
y = df['CO2 Emissions(g/km)']

print("\nShape of X (features):", X.shape)
print("Shape of y (target):", y.shape)
```

```
DataFrame loaded successfully. Displaying the first 5 rows:
  Make      Model Vehicle Class  Engine Size(L)  Cylinders Transmission \
```

0	ACURA	ILX	COMPACT	2.0	4	AS5
1	ACURA	ILX	COMPACT	2.4	4	M6
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6
4	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6

	Fuel Type	Fuel Consumption City (L/100 km)	\
0	Z	9.9	
1	Z	11.2	
2	Z	6.0	
3	Z	12.7	
4	Z	12.1	

	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	\
0	6.7	8.5	
1	7.7	9.6	
2	5.8	5.9	
3	9.1	11.1	
4	8.7	10.6	

	Fuel Consumption Comb (mpg)	CO2 Emissions(g/km)
0	33	196
1	29	221
2	48	136
3	25	255
4	27	244

Shape of X (features): (7385, 1)
 Shape of y (target): (7385,)

✓ Initial Linear Regression

Subtask:

Fit a simple linear regression model using 'Engine Size' to predict 'CO2 Emissions' and extract the initial coefficient (β_1).

Reasoning: I will import the `LinearRegression` model from `sklearn.linear_model`, instantiate it, fit the model using the prepared `X` and `y` data, and then extract and display the coefficient for 'Engine Size'.

```
from sklearn.linear_model import LinearRegression

# Instantiate a LinearRegression model object
model = LinearRegression()

# Fit the model to your data X and y
model.fit(X, y)
```

```
# Extract the coefficient ( $\beta_1$ ) for 'Engine Size'
beta_1 = model.coef_[0]

print(f"The initial coefficient ( $\beta_1$ ) for 'Engine Size' is: {beta_1:.4f}")

The initial coefficient ( $\beta_1$ ) for 'Engine Size' is: 36.7773
```

✓ Perform Bootstrap Resampling

Subtask:

Implement a bootstrap resampling process with $B=2000$ iterations. In each iteration, randomly sample the data with replacement, fit a linear regression model, and store the estimated β_1 coefficient.

Reasoning: To perform bootstrap resampling, I will set the number of iterations, initialize an empty list for coefficients, combine the feature and target into a single DataFrame, and then loop through the specified number of iterations to sample data, fit a linear regression model, and store the coefficient from each sample.

```
import numpy as np

# 1. Set the number of bootstrap iterations
B = 2000

# 2. Create an empty list to store the  $\beta_1$  coefficients
bootstrapped_coefficients = []

# 3. Create a combined DataFrame of X and y for easy sampling
data = pd.concat([X, y], axis=1)

# 4. Loop B times
for i in range(B):
    # a. Randomly sample rows from the combined data DataFrame with replacement
    # The sample size should be the same as the original dataset's size
    sample = data.sample(n=len(data), replace=True, random_state=i) # Using rar

    # b. Separate the resampled data back into feature X_sample and target y_sample
    X_sample = sample[['Engine Size(L)']]
    y_sample = sample['CO2 Emissions(g/km)']

    # c. Instantiate a new LinearRegression model
    model_bootstrap = LinearRegression()

    # d. Fit this new model to X_sample and y_sample
    model_bootstrap.fit(X_sample, y_sample)
```

```
# e. Extract the coefficient ( $\beta_1$ ) from the fitted model and append it to the
bootstrapped_coefficients.append(model_bootstrap.coef_[0])

print(f"Completed {B} bootstrap iterations. Stored {len(bootstrapped_coefficients)} coefficients."
print(f"First 5 bootstrapped coefficients: {bootstrapped_coefficients[:5]}")
```

Completed 2000 bootstrap iterations. Stored 2000 coefficients.
First 5 bootstrapped coefficients: [np.float64(36.72320825486899), np.float64(37.00000000000000), np.float64(36.99999999999999), np.float64(36.99999999999999), np.float64(36.99999999999999)]

✓ Calculate Confidence Intervals

Subtask:

Calculate the 90% and 95% confidence intervals for β_1 from the distribution of bootstrapped coefficients.

Reasoning: To calculate the confidence intervals, I will first convert the `bootstrapped_coefficients` list into a NumPy array. Then, I will use `np.percentile` to determine the lower and upper bounds for the 90% and 95% confidence intervals, respectively. Finally, I will print these values.

```
import numpy as np

# 1. Convert the bootstrapped_coefficients list into a NumPy array
bootstrapped_coefficients_array = np.array(bootstrapped_coefficients)

# 2. Calculate the 90% confidence interval
ci_90_lower = np.percentile(bootstrapped_coefficients_array, 5)
ci_90_upper = np.percentile(bootstrapped_coefficients_array, 95)

# 3. Calculate the 95% confidence interval
ci_95_lower = np.percentile(bootstrapped_coefficients_array, 2.5)
ci_95_upper = np.percentile(bootstrapped_coefficients_array, 97.5)

# 4. Print the calculated confidence intervals
print(f"90% Confidence Interval for  $\beta_1$ : ({ci_90_lower:.4f}, {ci_90_upper:.4f})")
print(f"95% Confidence Interval for  $\beta_1$ : ({ci_95_lower:.4f}, {ci_95_upper:.4f})")
```

90% Confidence Interval for β_1 : (36.2957, 37.3012)
95% Confidence Interval for β_1 : (36.2197, 37.3906)

✓ Visualize Bootstrapped Coefficients and CIs

Subtask:

Create a histogram of the bootstrapped β_1 coefficients and overlay the calculated 90% and 95% confidence intervals to visualize their spread and location. Ensure a legend is provided.

Reasoning: To visualize the bootstrapped coefficients and their confidence intervals, I will use `matplotlib.pyplot` to create a histogram and add vertical lines representing the 90% and 95% confidence intervals, along with appropriate labels and a legend.

```
import matplotlib.pyplot as plt

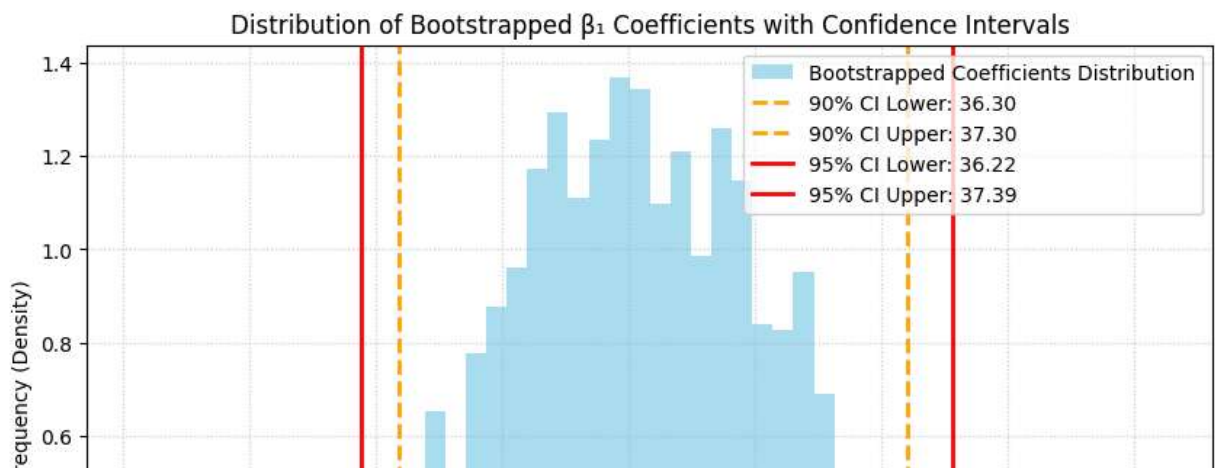
# Set up the plot
plt.figure(figsize=(10, 6))

# Create a histogram of the bootstrapped coefficients
plt.hist(bootstrapped_coefficients_array, bins=50, density=True, alpha=0.7, col

# Add vertical lines for the 90% confidence interval
plt.axvline(ci_90_lower, color='orange', linestyle='--', linewidth=2, label=f'90% CI Lower: 36.30')
plt.axvline(ci_90_upper, color='orange', linestyle='--', linewidth=2, label=f'90% CI Upper: 37.30')

# Add vertical lines for the 95% confidence interval
plt.axvline(ci_95_lower, color='red', linestyle='-', linewidth=2, label=f'95% CI Lower: 36.22')
plt.axvline(ci_95_upper, color='red', linestyle='-', linewidth=2, label=f'95% CI Upper: 37.39')

# Add labels and title
plt.xlabel('Bootstrapped  $\beta_1$  Coefficients')
plt.ylabel('Frequency (Density)')
plt.title('Distribution of Bootstrapped  $\beta_1$  Coefficients with Confidence Intervals')
plt.legend()
plt.grid(True, linestyle=':', alpha=0.6)
plt.show()
```



✓ Conclude on Significance

Subtask:

Based on the calculated confidence intervals, provide a conclusion on whether engine size significantly affects CO2 emissions.

Based on the calculated confidence intervals for the β_1 coefficient (the effect of 'Engine Size' on 'CO2 Emissions'):

- **90% Confidence Interval:** (36.2957, 37.3012)
- **95% Confidence Interval:** (36.2197, 37.3906)

Both the 90% and 95% confidence intervals do **not** include zero. Since the entire range of both confidence intervals is positive, it strongly suggests that the true coefficient for 'Engine Size' is not zero. This means that an increase in 'Engine Size' is associated with a statistically significant increase in 'CO2 Emissions'.

Therefore, we can conclude that **'Engine Size' significantly affects 'CO2 Emissions'**.

Final Task

Subtask:

Summarize the findings from the bootstrap analysis, including the estimated confidence intervals and the conclusion regarding the significance of engine size on CO2 emissions.

Summary:

Q&A

- **Significance of Engine Size on CO2 Emissions:** Based on the bootstrap analysis, 'Engine Size' significantly affects 'CO2 Emissions'. Both the 90% and 95% confidence intervals for the β_1 coefficient (the effect of 'Engine Size') do not include zero, and their entire range is positive, indicating a statistically significant positive relationship.

Data Analysis Key Findings

- The initial linear regression model revealed an estimated coefficient (β_1) for 'Engine Size' of approximately 36.7773. This suggests that for every one-unit

increase in 'Engine Size (L)', 'CO2 Emissions (g/km)' are predicted to increase by about 36.7773 units.

- A bootstrap resampling process with 2000 iterations was performed, yielding a distribution of β_1 coefficients.
- The calculated 90% confidence interval for the β_1 coefficient, based on the bootstrapped distribution, is (36.2957, 37.3012).
- The calculated 95% confidence interval for the β_1 coefficient is (36.2197, 37.3906).
- Both confidence intervals are entirely positive and do not include zero, providing strong evidence that 'Engine Size' has a statistically significant positive impact on 'CO2 Emissions'.
- The distribution of bootstrapped coefficients was visualized, clearly showing a concentrated distribution around the initial β_1 estimate and the bounds of the confidence intervals.

Insights or Next Steps

- The consistent positive and statistically significant relationship between 'Engine Size' and 'CO2 Emissions' indicates that vehicle engine size is a robust predictor of emissions. This information can be valuable for policymakers or manufacturers aiming to reduce CO2 emissions.
- Consider exploring other vehicle specifications (e.g., number of cylinders, transmission type) using similar bootstrap methods to identify additional significant factors or to build a more comprehensive predictive model for CO2 emissions.